

Sub-pixel Discrimination of Soil and Crop in Drone-based Hyperspectral Imagery

Manohar Kumar C. V. S. S.

Department of Earth and Space
Sciences

Indian Institute of Space Science and
Technology

Thiruvananthapuram, Kerala, India

manohar.iist@gmail.com

Rama Rao Nidamanuri

Department of Earth and Space
Sciences

Indian Institute of Space Science and
Technology

Thiruvananthapuram, Kerala, India

rao@iist.ac.in

Vinay Kumar Dadhwal

School of Natural Sciences and
Engineering

National Institute of Advanced Studies
Thiruvananthapuram, Kerala, India

dadhwalvk@hotmail.com

Abstract— Adapting technological interventions for plant-level nutrition optimization and pest-disease control requires detection and spatial aggregation of agricultural plant material—leaves, canopy gaps, and soil exposure. The potential of within-plant canopy crop-soil discrimination with sensitivity to canopy gaps and structure is not yet reported. This research aims to discriminate crop-soil within-plant canopies for plant-level discrimination using high-resolution hyperspectral imagery. A drone-mounted frame-based hyperspectral imaging sensor in the VNIR area is used for acquiring hyperspectral images of vegetable crops over an experimental farm in Bangalore, India. Further, reference in situ spectral reflectance measurements at plant-level of background soil, and crops are recorded from hyperspectral spectroradiometer and are used for generating a spectral library. Spectral signatures from the library are used as endmembers to simulate within-plant canopy level discrimination of plant and soil. Abundances of crop and soil are retrieved using various linear, non-linear, and sparse-based spectral unmixing approaches. The results are validated using ground truth fractional measurements carried out over diverse plant development and lighting conditions. The comparison of estimated and measured abundances indicates that plant-level soil and discrimination is possible with about 97% accuracy.

Keywords—Crop, Soil, Hyperspectral Remote Sensing, Spectral Library, Spectral Unmixing

I. INTRODUCTION

Various spectral imaging methods have been investigated for crop monitoring, detection, identification, phenology, and discrimination [1]. More recent efforts have been made to detect and identify crops at the plant level, expanding the scope of these technologies into the expanding realm of small-field farming [2]. Adapting technological interventions for plant-level optimization of nutrients and pest-disease control requires detecting and spatial aggregating agricultural plant material, such as leaves, canopy gaps, and soil exposure.

Large-scale crop mapping and characterization within the field variability for machine-initiated intervention has used high-resolution multispectral remote sensing [3]. Due to their ability to acquire image data down to the plant level, drones have quickly become important remote sensing devices. Drones equipped with the right spectrum imaging sensors might be invaluable tools for identifying and mapping agricultural features at the plant canopy level [4], [5]. Creating field- or patch-level automated agricultural sensing and monitoring systems relies heavily on automatic crop-soil differentiation at the plant level. Several studies reporting on crop-soil discrimination have used high-resolution remote sensing data [5]–[8]. The remote sensing use of analyst-assisted, pixel-level discrimination of crop and soil is well-explored and proven, thanks to spectral index-based

differentiation [6]. However, the potential of canopy gaps and structures to differentiate crop and soil at the plant level has not been observed. Several areas of precision agriculture may benefit from automated sub-pixel crop-soil differentiation. Theoretically, spectral unmixing [9] provides a straightforward method for distinguishing agricultural soils at the sub-pixel level. Hyperspectral imaging captured by drones makes it possible to assess the feasibility of fine-scale crop and soil differentiation. This study aims to investigate crop-soil discrimination at the level of the plant canopy. The areal distribution of the crop and soil was estimated using various linear, non-linear, and sparse-based spectral mixture models to validate the discrimination between the crop and soil.

II. STUDY AREA, DATA ACQUISITION, AND METHODS

A. Study Area

The experimental setting is situated in the fields of the University of Agricultural Sciences, Gandhi Krishi Vignan Kendra, which is geographically positioned at coordinates 13°5.262'N 77°33.982'E in Bangalore, Karnataka. The cabbage crop is chosen for soil discrimination based on comprehensive cultivation and consumption trends.

B. Data Acquisition and pre-processing

The drone hyperspectral imagery was obtained using a compact hyperspectral imaging sensor (Cubert Fireflye S 185). The imager employs a hyperspectral sensor comprising 139 spectral channels, each with a sample interval of 4nm, covering the electromagnetic spectrum ranging from 450nm to 950nm. The sensor was positioned on a quadcopter drone, and data was collected at an altitude of 40 meters. The radiance imagery captured is subjected to normalization using the white reference panel. A pre-processed true colour composite of the hyperspectral imagery is shown in Figure 1.

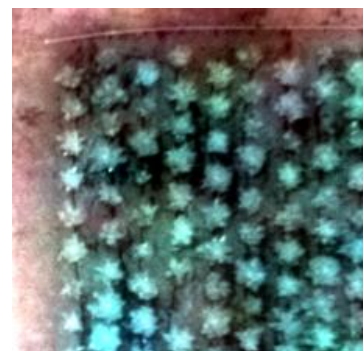


Figure 1. True colour composite of drone hyperspectral imagery taken at an altitude of 40m.

C. Field or In-situ Spectral Library

In conjunction with the image-based data, we obtained in-situ or field reflectance spectral measurements at various locations inside the cabbage crop field, including both the cabbage plants and the soil. This study used a ground-based hyperspectral spectroradiometer (SVC, HR1024i model). The equipment can measure reflectance throughout the wavelength range of 350 to 2500nm, exhibiting a spectral resolution of 3nm. A distance of 50 cm is maintained above the cabbage plant, with the lens having a field of vision of 4°. The acquisition of reflectance spectrum data was conducted in parallel with using a white reference panel coated with barium sulphate (BaSO₄). This was done in order to calibrate the radiance spectral measurements. The spectral measurements of field reflectance were resampled to align with the spectral range and bandwidth of both ground-based and drone-based hyperspectral imagery. Figure 2 shows the reflectance profiles of cabbage and soil that were obtained using a field spectroradiometer.

D. Methodology

The pre-processed hyperspectral imagery and the prepared independent spectral library were utilized in the spectral unmixing modeling using nine distinct spectral unmixing models. These models can be categorized into linear methods (CLS and FCLS), non-linear methods (GBM, Fan-BL, PPNM, MLMM, and Hapke), and sparse methods (SUnSAL and CLSUnSAL) [10]–[14]. Figure 3 provides a comprehensive depiction of the methodological process flow. The validation of the areal abundances of vegetable crops was conducted by comparing the estimates obtained from unmixing approaches with actual ground areal percentage calculations. Additionally, the statistical measure spatial-root mean square error (S-RMSE) and overall root mean square error (O-RMSE) [15], [16] was computed to assess the similarity between the input airborne hyperspectral imagery and the imagery generated from the spectral library. The spectral unmixing quality evaluation often uses the signal-to-reconstruction error (SRE) as a validation parameter, which was also calculated in this study.

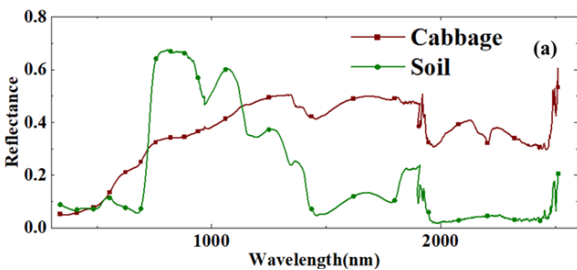


Figure 2. Spectral signature of a cabbage plant and soil acquired using field spectroradiometer.

III. RESULTS AND ANALYSIS

The crop and soil discrimination in the drone hyperspectral imagery through various spectral unmixing algorithms and the statistical error analysis are presented in the following sub-sections.

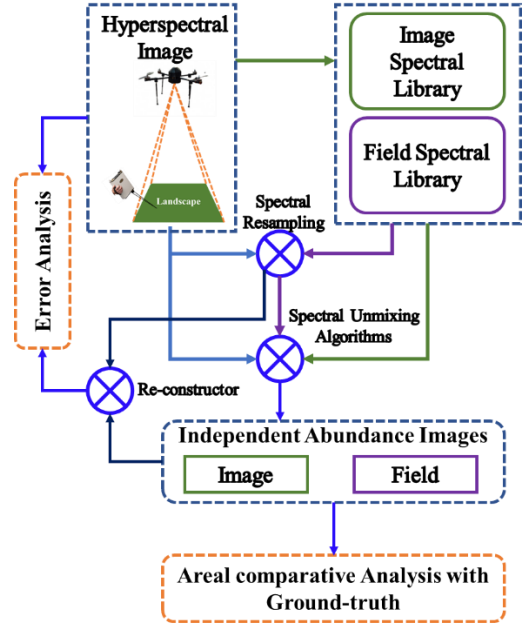


Figure 3. The procedure used to discriminate the crop and soil at plant level from acquisition to areal distribution cover and statistical error comparison with the prepared ground truth.

A. Crop and soil discrimination

Figure 3 displays the estimated abundances of the crop and soil at the plant level in the drone hyperspectral imagery, as obtained using several spectral unmixing algorithms using two different spectral library sources. The areal statistics are shown in Table I, whereas Table II displays the mean pixel error at each plant level based on ground truth locations.

The best possible discrimination of cabbage crop and soil is possible in the drone hyperspectral imagery at 98% (CLS) and 97% (FAN_BL), when the spectral library is from the drone imagery, and 85% (FCLS) and 99% (Hapke) discrimination is possible using the field spectral library as a source, respectively.

The origin of endmembers is a significant determinant of the accuracy of crop and soil discrimination at the plant-level. The performance of the drone based imagery spectral library is superior when compared to the field spectral library. The improved performance of image-based endmember spectra may be attributed to the potential presence of a cohesive blend of endmembers inside crop and other sub-pixel level land coverings in the imagery. Disregarding the relative comparison, the efficacy of estimating abundance for cabbage crop at plant level using external reference spectra seems to be suboptimal, but the opposite holds true for soil samples.

In the context of cabbage crop analysis, it has been shown that linear algorithms exhibit superior performance in comparison to non-linear spectral unmixing algorithms. Conversely, when considering soil analysis, the converse holds true, with non-linear algorithms demonstrating greater performance. The incorporation of various interactions inside algorithms has the potential to diminish the overall performance. Further investigation is required to quantify numerous interactions.

B. Error analysis

Figure 4 represents S-RMSE, and Table III provides the values of the Overall RMSE, and SRE computed between the

input hyperspectral imagery and the reconstructed hyperspectral imagery using two different imagery and field spectral libraries for various spectral inversion techniques. The average error in geographical location for plant-level discrimination is shown in Table II.

The results shown in Table II demonstrate that the discriminating of cabbage crops at the plant level may be effectively achieved using the imaging spectral library, in contrast to the field spectral library. The Hapke model is shown to be unsuitable for modelling cabbage crop at the plant level, however it is inversely applicable for soil analysis. This assertion is further supported by the evidence from SRE values shown in Table III.

Table I. The areal abundance of cabbage crop and soil using two different endmember sources.

	Field spectral library		Image spectral library	
	Crop	Soil	Crop	Soil
CLS	58.532	132.095	98.335	89.332
SUnSAL	58.533	132.094	98.335	89.331
FCLS	85.123	84.633	91.14	96.185
CLSUnSAL	85.123	84.633	91.14	96.185
FAN_BL	69.459	86.309	81.738	97.403
GBM	73.801	86.342	89.541	96.178
PPNM	25.815	99.927	86.881	90.851
MLMM	58.77	99.099	92.047	96.422
HAPKE	5.351	99.906	77.308	84.612

Table II. Mean S-RMSE of crop and soil at respective locations using field and imagery spectral libraries.

	Field spectral library		Image spectral library	
	Crop	Soil	Crop	Soil
CLS	88.11	0.38	9.558	3.965
SUnSAL	88.107	0.38	9.557	3.965
FCLS	14.877	15.367	8.86	3.815
CLSUnSAL	14.877	15.367	8.86	3.815
FAN_BL	30.541	13.691	18.262	2.597
GBM	26.199	13.658	10.459	3.822
PPNM	74.185	0.073	13.119	9.149
MLMM	41.23	0.901	7.953	3.578
HAPKE	94.649	0.094	22.692	15.388

IV. CONCLUSION

This research aims to expand the use of a spectral unmixing technique for distinguishing cabbage crop and soil at the individual plant level in very high spatial resolution drone hyperspectral imagery. The effective differentiation of the geographical distribution of cabbage crops and soil at the plant level is achievable in agricultural fields. The

performance of the drone imaging spectral library is superior to that of the field spectral library for the cabbage crop, achieving an accuracy rate of 98.3%. Similarly, the field spectral library has the highest performance for the soil, with an accuracy rate of 99.9%. The efficacy of the linear unmixing model has been seen in the context of cabbage crop analysis, whereas non-linear unmixing models have been shown to be more suitable for soil analysis.

Table III. Overall RMSE and SRE computed between the original input and reconstructed hyperspectral imagery using image and field spectral libraries

	Field spectral library		Image spectral library	
	O-RMSE	SRE	O- RMSE	SRE
CLS	0.0280	21.7751	0.0144	27.5374
SUnSAL	0.0280	21.7751	0.0144	27.5373
FCLS	0.0737	13.3679	0.0822	13.2870
CLSUnSAL	0.0737	13.3679	0.0822	13.2870
FAN_BL	0.0693	13.9036	0.0853	13.0175
GBM	0.0692	13.9186	0.0818	13.3600
PPNM	0.0368	19.4020	0.0210	24.3063
MLMM	0.0243	23.0158	0.0210	24.2230
HAPKE	0.1178	9.2959	0.1122	10.9839

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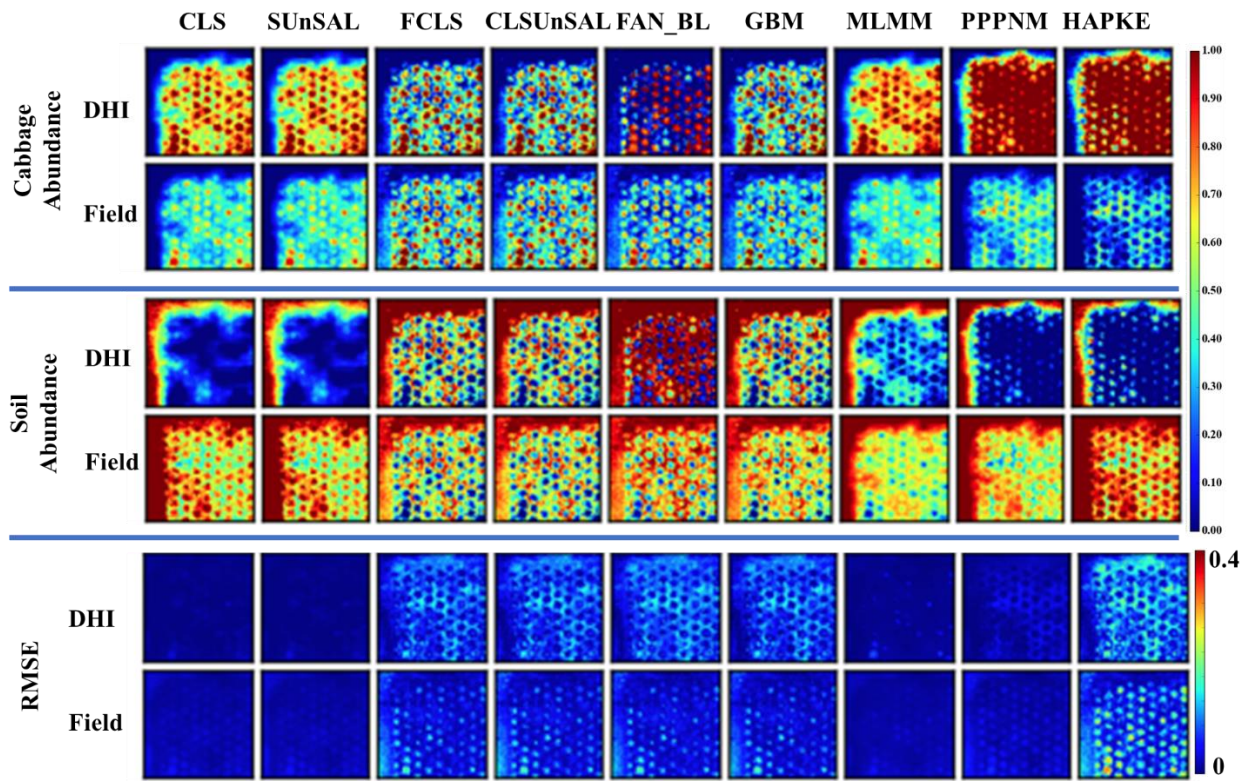


Figure 4. Estimated cabbage and soil abundances using spectral libraries (Field and DHI) for various spectral unmixing algorithms along with pixel-wise RMSE.

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